

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

8th September 2021

Subject SP6 – Financial Derivatives

Time allowed: 3 Hours 30 Minutes (14.30 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. *Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.***
- 2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.***
- 3. *Mark allocations are shown in brackets.***

- Q. 1)** You are required to advise a company which is considering selling a 4-year credit default swap. Initial estimate of default for the company (who issued the bond underlying the CDS) is 4% for first year but increases by 0.5% each year thereafter. You also assume a recovery rate of 40%, and that any default will take place at the mid-point of the year. The risk-free interest rate is 12%, continuously compounded.
- i) The Sales department, to maintain the customer relationship, assured the buyer that the pricing of CDS would assume no default in between year 2 – 3. What would be your theoretical estimate of the CDS using the given information. (Assume premium paid annually in arrears) (8)
- ii) In your opinion, how much loss would the company be making due to this deal (in spread terms)? (6)
- [14]

- Q. 2)** As the treasurer of a bank, you have decided to use the Black's model to value vanilla interest rate options (caps, floors and swaptions). The bank is looking to extend its range of products into more exotic interest-rate swaps and options denominated in Euro and you need to build a term structure yield curve model for pricing and hedging these instruments.
- i) Given this information, answer the following questions:
- a) Outline the reasons for choosing a full yield curve model in this situation. (3)
- b) Set out the main criteria such a model must satisfy. (3)
- c) Explain the relevance of a "no arbitrage" condition in this context. (3)

The Hull-White one-factor yield curve model projects the short-term rate r according to the formula:

$$dr = a(t)(b(t) - r)dt + \sigma(t)dz$$

where $\sigma(t)$ is the time-dependent short-rate volatility and z is a standard Brownian motion. The bank has decided to use an implementation that sets parameters $a(t)$ and $\sigma(t)$ as constants a and σ respectively for all t .

- ii) Describe how you would construct a trinomial rate tree for this model, calibrated to the term structure of zero-coupon bond prices and caplet volatilities. You do not need to derive the branching equations. (5)
- iii) Discuss the extent to which your Hull-White implementation will fulfil the desirable criteria you identified in (i)(b). (5)
- [19]

- Q. 3)** Indiglo Airlines plans to order 100,000 barrels of Aviation Turbine Fuel (ATF) in three months' time. It is concerned about the risk that fluctuating fuel prices pose to its financial

performance. In this context, the Chief Financial Officer of the airlines has decided to hedge this risk using ATF futures, for which a contract is available which expires in three months' time.

You are given the following information:

- The spot price of one barrel of ATF is \$101.20.
- The relevant futures price for delivery of one barrel of ATF is \$106.30.
- The standard deviation of changes in the spot price of a barrel of rocket fuel over the next three months is \$4.20.
- The standard deviation of changes in the relevant futures price for ATF over the next three months is \$2.50.
- The coefficient of correlation between changes in ATF futures prices and rocket fuel spot prices is 0.75.
- One ATF future represents 1,000 barrels of ATF.

i) Define the following terms:

- a) Cross hedging (1)
- b) Hedge ratio (1)
- c) Minimum variance hedge ratio (1)
- d) Tailing the hedge (1)

ii) Calculate the number of ATF futures that Indiglo should buy to minimize the variance of the project's total cost. (2)

iii) Discuss in general terms what can be deduced about ATF storage costs from ATF futures prices. (4)

[10]

Q. 4) An investment firm plans to structure a total return swap with the following assets:

- Asset A: 30 years amortizing bond with 5% effective discount rate and principal of INR 100m with redemption value of 0
- Asset B: 30 years adjustable-rate amortizing bond with initial rate of 3% effective discount rate, which would get adjusted to 5.25% in 3-year period and principal of INR 100m with redemption value of 0

Assume that all rates are annually compounded.

- i) What is a total return swap (TRS)? What is its advantage? What is the value of TRS at start of the contract? (2)
- ii) Derive the formula to estimate the annual payments for the amortizing bond. (2)
- iii) Estimate the pay-out of the TRS at year 4 assuming investment firm owns Asset A and has swapped the returns with Asset B. (5)
- [9]

Q. 5) An investment firm uses the binomial model for the pricing of derivatives:

- i) Describe the binomial option pricing formula in binomial model of derivative valuation. How does risk premium for the underlying asset impact the formula? (2)
- ii) Discuss what each of the following represents in the binomial model? (2)
- The difference between the up and down factors.
 - The ratio of the underlying value to the exercise price.
- iii) Discuss which of the following are not a factor in pricing a call option in the binomial model? (2)
- The risk-free rate
 - The exercise price
 - The probability that the underlying will go up

Assuming that a stock price follows the following process:

$$dS = \mu S dt + \sigma S dW_t$$

where μ , σ are positive constants and W_t is a standard Brownian motion. The continuously compounded risk-free interest rate r is constant.

- iv) Estimate the value of American call option with spot price of 100 and strike price of 105 using a 2-step recombining binomial tree. You can assume for small t the upward multiplier $u = \exp(\sigma \sqrt{\Delta t})$
- $\sigma = 20\%$
 - $r = 5\%$ continuously compounding

State any assumption in the calculation. (4)

[10]

- Q. 6)** i) In a forward contract what is the difference between the forward price and value of the forward contract. State the formula for forward price of asset assuming underlying asset generate cash payment (c) and incur the storage cost (s), spot price of S and interest rate of r . All information as at start of the contract.

(3)

ii) Calculate the value of oil forward contract where Spot = 1000, $r = 5\%$, storage cost = 3% , duration 90 days. The oil can also be used as collateral for 2% return. All rates are continuous compounding (2)

iii) What is a forward rate agreement (FRA)? Provide the cashflows for a 30-day FRA with a fixed rate (the FRA rate) where the underlying is 90-day Mibor. How can you create a synthetic FRA to replicate the payoff of this FRA? (3)

iv) A company requires to make a payment in USD after 6 month and is keen to estimate the expected Fx rate at that point. Derive a formula forward exchange rate (USD/INR) in 6 months' using arbitrage free approach to value forward exchange rate contract given interest rate in India (R_{ind}) and US (R_{usd}). (3)

v) Estimate the Fx rate based on the below information

- Spot Fx (USD/INR) 0.013
- R_{ind} 6%
- R_{usd} 3%

(2)

[13]

Q. 7) An insurance company aims to determine the impact of interest rate on the portfolio of fixed income securities by describing the curve of interest rate by three factors - level, steepness and curvature.

i) Describe how the interest rate changes for various duration in following changes

- Up and down in level.
- Increase and decrease of steepness.
- Increase and decrease of curvature.

Describe how a combination of above 3 movement can be used to describe the overall movement of the yield curve. (5)

ii) Suggest a linear model to describe the percentage change in portfolio value with change in the above components. (2)

iii) Suggest approach to estimate the parameters for the model and identify potential limitation of the model. (2)

[9]

Q. 8) i) State usability of effective duration and convexity over modified duration and convexity of bonds. State how effective duration and convexity is calculated. (3)

ii) What would be the impact on effective duration if bonds have

- Call option

- Put option

(2)

iii) State and explain for callable bond how convexity would vary based on level of yield curve.

(2)

[7]

Q. 9) An exchange traded fund (ETF) is a type of security that tracks an index, sector, commodity, or any other asset, but which can be purchased or sold on a stock exchange the same as a regular stock. ABCD Asset Management Company Ltd. has recently launched a 2:1 leveraged ETF that promises to deliver twice the returns of the NIFTY index.

i) What are the ways in which the ETF can deliver the promised returns?

(2)

ii) Which segment of investors do you think would find this product attractive?

(2)

iii) What are the advantages and disadvantages of such a product?

(5)

[9]
